A050 – The Recursion: Non-Closure and Logarithmic Spiral

Johann Pascher

July 2026

Contents

.1 The Recursion

[STIPULATION] The scale structure of the theory is generated by the map

$$\xi_{n+1} = \xi_n \left(1 - 100\, \xi_n\right)$$

starting from $\xi_0 = 4/30000$.

This is a damped feedback: the factor $(1 - 100\xi_n)$ is less than one, the sequence falls monotonically. In communications engineering terms it is a loop with gain below one – it contracts, it does not oscillate. The gain is not adjustable: it depends solely on ξ, which is stipulated. There is no tuning knob in this loop.

.2 The Non-Closure Theorem

[K] The recursion has exactly one fixed point, namely zero. Every value $c \in (0, 1)$ that the sequence assumes is a transit point, not a target.

The proof is elementary: from $\xi = \xi(1 - 100\xi)$ it follows that $100\xi^2 = 0$, hence $\xi = 0$. And because the sequence is strictly monotonically decreasing but stays positive, it will not revisit any of its values in finitely many steps and will not reach zero in any finite step.

[K] From this follows a statement that is easy to overlook and yet load-bearing: *from the recursion alone, no individual numerical value can be singled out*. For every target c there exists a depth $k^*(c)$ at which the sequence comes close to it – equally so for every c, with identical parameter status. One who tries to justify a number by claiming the recursion passes through it has justified nothing.

[K] This is a result, not a gap. It closes an entire class of derivation attempts: the question of whether a specific constant can be obtained from the recursion depth is *answered*, and the answer is no. What remains open is only whether there is an anchor outside the recursion.

.3 The Frozen Winding

[B] Holding ξ fixed instead of letting it run, the winding closes after exactly

$$\frac{1}{100\,\xi_0} = 75$$

circuits. The rotation number is then $74/75$, and the missing piece is the closure comma of A040.

This is the *rolled-up*, compact reading: one scale, one closed winding, an integer number of circuits.

.4 The Unrolled Reading: A Logarithmic Spiral

[K] Letting ξ run, the sequence decays asymptotically as

$$\xi_n \approx \frac{1}{100\,(n+75)},$$

and the cumulative deficit becomes a harmonic sum, hence logarithmic:

$$D(k) \approx \ln\!\left(1 + \frac{k}{75}\right).$$

[K] Setting $\rho = k + 75$ and $\beta = 2\pi D$, we get

$$\rho = 75\, e^{\beta/2\pi},$$

a logarithmic spiral with self-similarity ratio e.

Two points, both important.

The ratio e is *not an additional parameter*. It is the base of the natural logarithm and comes from the harmonic sum – it is not chosen, it falls out. The pole at $n = -75 = -1/(100\xi_0)$ is likewise not fitted but fixed by ξ_0.

The scale invariance is *continuous*, not discrete log-periodic. There are no distinguished steps, no preferred multiples.

[B] Rolled up and unrolled are two readings of the same object, not two objects: the closed 75-winding is the compact single-scale view, the spiral the unrolled across-scales view.

.5 Why the Recursion Does Not Belong on a Single Axis

[B] The formulation that time rolls up suggests that the time axis is distinguished. It is not. Through $\tilde{T} \cdot m = 1$, every statement about time has a mirror image about mass: if time advances by K_{frak} per step, mass advances by $1/K_{\text{frak}}$. The deficit is the same $100\xi_n$ on both sides.

[B] The factor 100 therefore does not migrate between the axes – it is mirrored. Both projections lead to the same ratio e. The assignment to time is a convention, not a distinction.

.6 Verification Scripts

- Fixed-point theorem and non-closure: unique fixed point 0, monotonicity, and the construction of $k^*(c)$ for arbitrary targets – as evidence that no value is distinguished: `python/A_Serie_Skripte/a050_nichtschliessung.py`

- Spiral ratio: numerical determination of the ratio from the cumulative deficit, comparison with e, and check for continuous rather than log-periodic scale invariance: `python/A_Serie_Skripte/a050_spirale_e.py`

.7 Open Questions

First, the recursion depth is not fixed. The theory gives no rule at which k a physical quantity is to be read off. By the non-closure theorem it cannot – this is not a negligence, but proved.

Second, this implies a hard limit for all fine values. Every quantity whose justification would amount to the recursion reaching it is thereby *not* justified. This applies notably to the fine value of the refraction strength δ; it remains open and is booked in A230.

Third, the recursion uses the factor 100, which is derived in A040 (cumulative RG run, consistency condition; A040). It is used here, not re-derived. The non-closure holds independently of this – it follows from the form of the map, not from the numerical value – but the concrete number 75 does not.

.8 Supersedes

This document subsumes: Dok. 203 (R-operator), Dok. 275 (ξ-recursion), Dok. 295 (time as logarithmic spiral). Incorporated: P37 (three roles of φ separated), P35 (triviality of the power representation).

.9 Use of AI Tools

This document was created with the assistance of AI tools. Responsibility for content and errors rests entirely with the author. Full wording of the rules in A010.